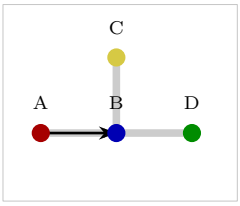
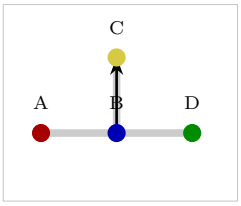
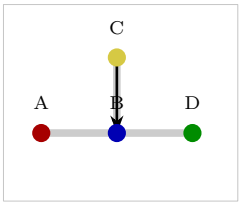
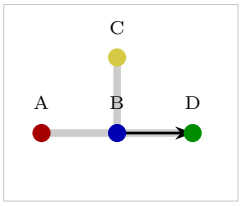
Move a pawn on the roads. Keep each journey as ordered step tiles. Two adjacent steps along the same road in opposite directions are an out-and-back pair. Remove only out-and-back pairs, and replay the other steps in order. The map, start, and finish stay fixed.

input



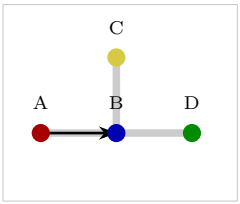


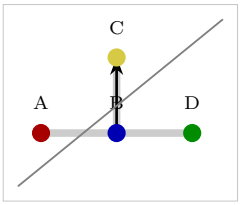


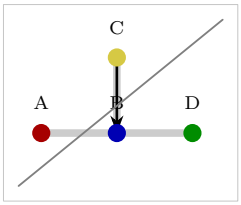


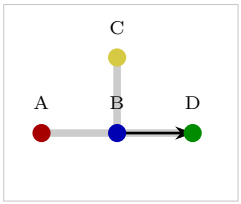
$A \to B \to C \to B \to D$

remove

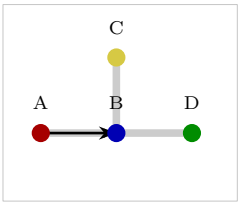


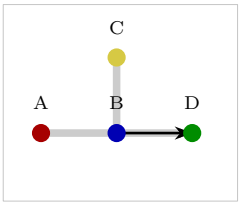






output





$A \to B \to D$

No steps left means the pawn stays at its start.

Problem 1: Shorten each recorded trip until no move is possible. Make a different seven-step trip from C to F that finishes with the same route as the last one.

$A \to B \to A \to B \to D \to F \to D \to E$

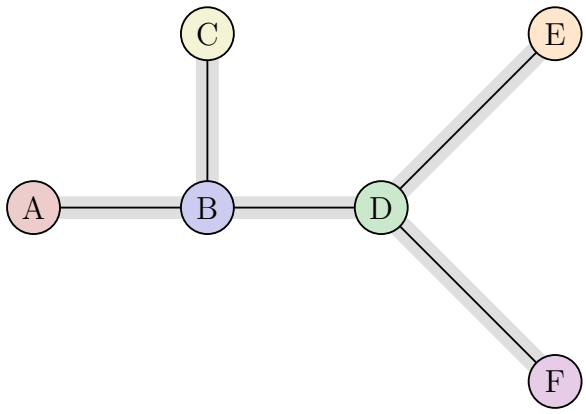
$\longrightarrow$ _____

$A \to B \to D \to F \to D \to E$

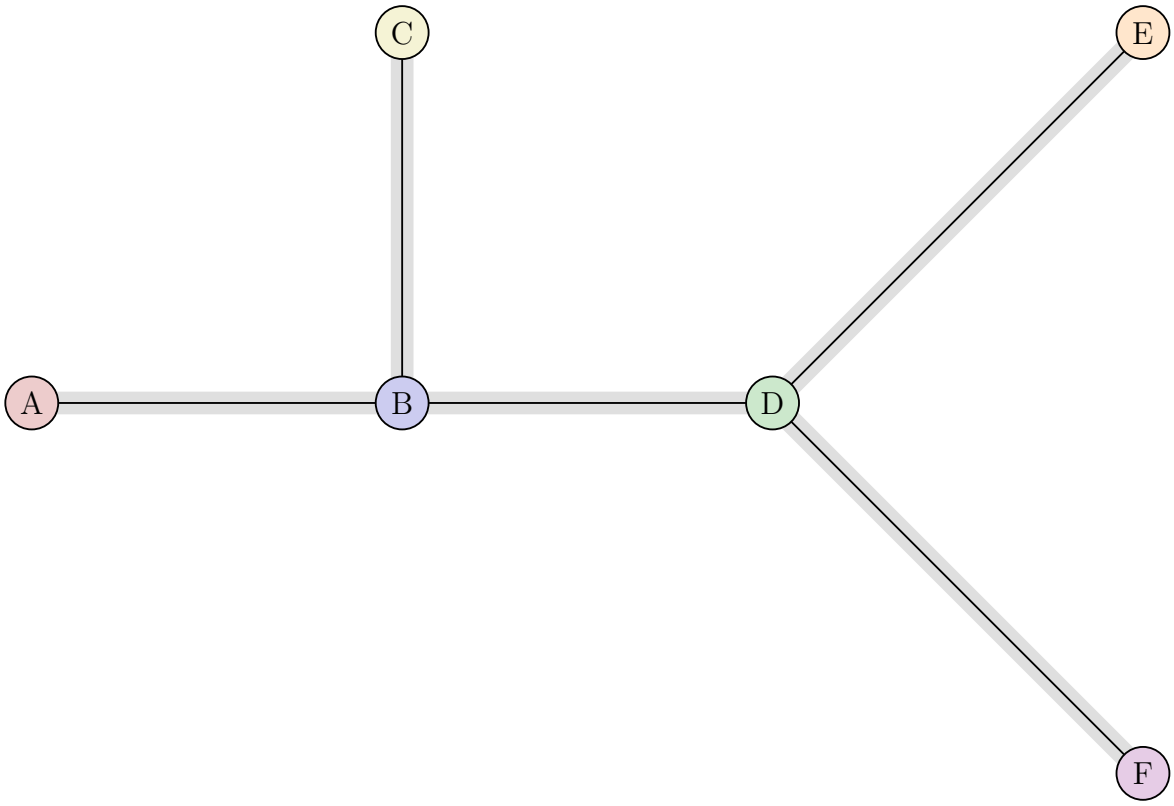
$\longrightarrow$ _____

$C \to B \to C \to B \to D \to F$

$\longrightarrow$ _____

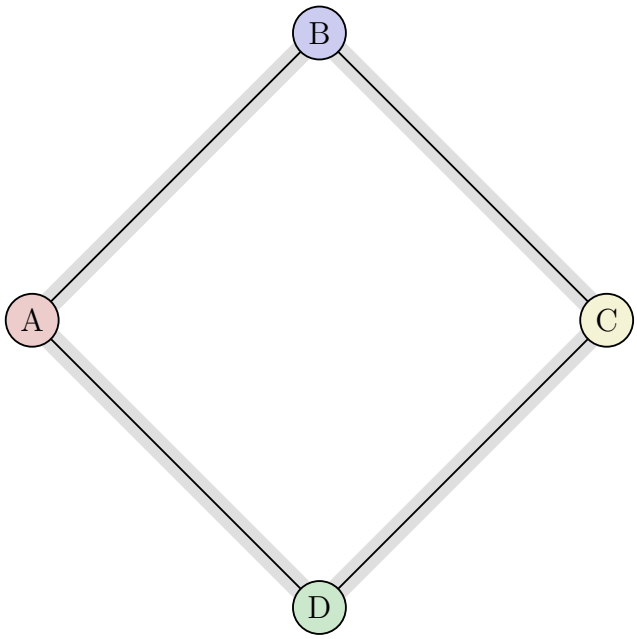


Problem 2: On this map, can a trip from A back to A survive after every out-and-back pair is removed? Explain why your answer holds for trips of any length.



Problem 3: Can two trips from C to F finish with different routes after every out-and-back pair is removed? Explain using the roads.

Problem 4: Shorten each recorded trip in different orders. Can the choice of first out-and-back pair ever change the final route on any road map? Explain, or give a counterexample.



$A \to B \to A \to B \to C \to B \to A \to D \to C \to D \to A \quad \longrightarrow$ _____

$A \to B \to A \to D \to C \to B \to C \quad \longrightarrow$ _____

$A \to B \to C \to B \to C \to D \to A \to B \to A \quad \longrightarrow$ _____